

Mitigation Economics

From Stylised Facts to Policy Design

Working Draft

1. Table 1: National-Level Policy Framework

	Assumption / stylised fact	Model in words	Mathematical structure	Optimal policy	Deviation from Pigouvian	Practical implication
<i>Baseline: the textbook case</i>						
0	Single externality, rational agents, no other distortions	Firms and consumers optimise perfectly; the only problem is unpriced CO ₂ .	Social cost = private cost + $\tau \cdot e_f$. Agents minimise total cost.	Pigouvian carbon tax τ = SCC. Nothing else needed.	None. This is the benchmark.	Almost never holds in practice. Useful only as a reference point.
<i>Deviations: adding realism one stylised fact at a time</i>						
1	Putty-clay capital. Fuel choice is made at investment; existing stock is locked in.	Substitution only at the investment margin. Aggregate fuel mix changes only as capital stock turns over.	Vintage capital model. New investment: CES over current prices. Stock: $K_{f,v,t} = K_{f,v,v}(1 - \delta)^{t-v}$. Aggregate: $w_f = \sum_v K_{f,v} / \sum_{f,v} K_{f,v}$.	Same τ = SCC, but the speed of adjustment is slow. Need a credible long-term price path, not just today's price.	Carbon price must be <i>expected to persist</i> over the asset lifetime. A temporary price is much less effective than a permanent one per \$/tCO ₂ .	Favours durable instruments (standards, long-dated CfDs) over instruments vulnerable to political reversal.
2	Behavioural / myopic consumers. Households use high implicit discount rate ($r_c \gg r_s$).	Consumers undervalue future fuel savings relative to upfront cost. Underinvest in efficient/clean capital even with correct carbon price.	Consumer evaluates: $p^{\text{total}} = k_{fs} \cdot a(r_c) + \hat{p}_{fs}$ with $r_c = 20\text{--}30\%$ vs social $r_s = 3\text{--}5\%$. CES substitution responds to p^{total} at consumer r_c .	Carbon tax plus technology standard . Tax corrects externality; standard corrects behavioural wedge. Two failures → two instruments (Tinbergen).	Carbon tax alone is insufficient. Even at the “right” τ , consumers don't adopt enough clean technology because they over-discount fuel savings.	Justifies vehicle emission standards, appliance MEPS, building codes <i>alongside</i> carbon pricing. These are complements, not substitutes.
3	Learning-by-doing / knowledge spillovers. Deploying clean tech today reduces its future cost; deployer doesn't capture full benefit.	Two externalities: emissions and knowledge. Carbon tax corrects the first; deployment subsidy corrects the second.	Learning curve: $c_t = c_0 \cdot (\sum_{\tau < t} Q_\tau)^{-\beta}$. Social value of deployment exceeds private value by the spillover $\partial c / \partial Q$ · (future deployments).	Carbon tax plus deployment subsidy (FiT, auction, production tax credit). Subsidy = shadow value of the learning externality.	Carbon tax alone produces too little early deployment. The knowledge benefit to future adopters is an unpriced positive externality.	Justifies RE subsidies, EV incentives, green H ₂ support — especially for technologies pre-tipping-point. Subsidies can be temporary (until tipping point crossed).
4	Capital market failures. Clean investment faces higher financing costs (novelty risk, small scale, sovereign risk).	Investors demand a risk premium on clean relative to incumbent. Raises effective clean cost above true social cost.	$p^{\text{total}} = k_{fs} \cdot a(r_{\text{market}}) + \hat{p}_{fs}$ where $r_{\text{market}} = r_{\text{true}} + \rho_{\text{risk}}$. Risk premium ρ inflates clean capital cost.	Carbon tax plus derisking (guarantees, concessional finance). Derisking = reduce ρ so that $r_{\text{market}} \rightarrow r_{\text{true}}$.	Carbon tax alone is insufficient because the price signal is dampened by the financing wedge. \$50/tCO ₂ carbon tax “feels like” \$25/tCO ₂ when WACC is double.	Justifies MDB guarantees, concessional lending, blended finance. Highest leverage when risk premium is largest (developing countries, novel technologies).
5	Network externalities / coordination failures. Clean adoption requires complementary infrastructure (charging, grids, H ₂ pipelines).	Individual incentives insufficient: everyone waits for the network; nobody builds the network. Chicken-and-egg.	Coordination game with multiple equilibria. Low-adoption equilibrium is stable. Price signal alone doesn't shift to high-adoption equilibrium.	Carbon tax plus public infrastructure investment or coordination mandates .	No carbon price, however high, solves a coordination failure. You need a “big push” on infrastructure to shift equilibrium.	Justifies public EV charging buildout, grid expansion, H ₂ infrastructure commitments. These are complements to carbon pricing, not substitutes.

	Assumption / stylised fact	Model in words	Mathematical structure	Optimal policy	Deviation from Pigouvian	Practical implication
6	Policy reversal risk. Carbon price may be repealed after next election.	Investors discount future carbon price by probability of reversal. Effective signal at investment decision $\ll$ nominal price.	Effective $\tau^{\text{eff}} = \tau \cdot \text{PV}(\text{survival probability over instruments})$. If 30% reversal risk, $\tau^{\text{eff}} \approx 0.5\text{--}0.7 \cdot \tau$.	Choose durable instruments first. Standards and mandates are harder to repeal than explicit prices. CfDs and long-term contracts lock in the signal.	The “optimal” carbon tax loses much of its value if it’s not credible. A “suboptimal” standard at lower implicit \$/tCO ₂ can outperform if it’s more credible.	Explains why vehicle standards (implicit \$100–200/tCO ₂ , highly durable) deliver more abatement than carbon taxes (explicit \$30–50, reversal-prone). Favours embedding price signals in long-term contracts (CfDs).
7	Political economy: concentrated losers, diffuse winners. Carbon pricing creates visible cost increases for identifiable groups.	Losers organise politically; winners don’t. Carbon tax faces opposition proportional to its visibility and regressivity.	Political constraint: max feasible τ is well below SCC. Constraint binds hardest for visible, broad-based instruments.	Use less visible instruments first. Embed carbon price in standards (hidden in product costs). Use revenue to compensate losers. Use feebates (revenue-neutral $\rightarrow$ no visible cost increase).	The first-best instrument (transparent Pigouvian tax) is the most politically constrained. Second-best instruments (standards, feebates) achieve more abatement because they’re feasible at higher implicit rates.	Explains the actual global policy portfolio: fuel excises and standards do most of the work; explicit carbon prices cover <5% of emissions at rates above \$50/tCO ₂ . The “suboptimal” portfolio is the realistic one.
8	Tipping points / self-reinforcing dynamics. Past a threshold, clean tech is permanently cheaper and adoption is self-sustaining.	Policy only needs to push through the tipping point. After that, the market takes over.	S-curve: learning + scale $\rightarrow$ cost crosses below fossil. Once crossed, positive feedback (more deployment $\rightarrow$ lower cost $\rightarrow$ more deployment).	Front-load support; withdraw after tipping point. Temporary subsidies/standards are optimal if they trigger permanent transition.	Permanent carbon tax is unnecessarily costly if the tipping point can be crossed with temporary support. But the carbon tax does no harm; the subsidy is more efficient at triggering the tip.	Solar and wind have crossed the tipping point. EVs are approaching it. Policy should now shift from deployment subsidies to infrastructure and integration support. Standards remain relevant for sectors pre-tipping-point (heat pumps, green H ₂).

2. Table 2: Global / International Framework

	Assumption / stylised fact	Model in words	Mathematical structure	Optimal international instrument	Deviation from national-only optimum	Practical implication
<i>Baseline: the global textbook case</i>						
0	Global externality, sovereign nations, no transfers	CO ₂ is a global public bad. Each country's optimal national carbon price is below the global optimum (free-riding).	Nash equilibrium: $\tau_i^{\text{Nash}} < \text{SCC}$ for all i . Cooperative optimum: $\tau^* = \text{SCC}$ everywhere.	Global uniform carbon tax at SCC, with side payments to ensure participation.	None. This is the global benchmark.	Has never existed and probably never will. Useful only as a reference point.
<i>Deviations: adding global realism</i>						
G1	Sovereign risk premium. Clean energy WACC in developing countries 2–3× higher than in advanced economies for same technology.	Same turbine, same resource, radically different LCOE. The risk premium — not the technology — is the binding constraint on clean deployment.	$\text{LCOE}_i = k \cdot a(r_i) + \text{O\&M}$. For $r_{\text{SSA}} = 14\%$ vs $r_{\text{EU}} = 5\%$: $a(r)$ roughly doubles. LCOE doubles despite same k .	MDB guarantees and concessional finance to reduce r_i in developing countries. Highest-leverage global instrument.	No national carbon price, however high, can compensate for a financing cost that doubles the LCOE. Even \$200/tCO ₂ can't make RE competitive at 15% WACC in some markets. International risk reduction is necessary.	\$1 of MDB guarantee capital avoids 10–20× more CO ₂ than \$1 of carbon credit. The case for MDB capital adequacy reform is the strongest in all of climate economics.
G2	Global technology learning. Deployment anywhere reduces costs everywhere (knowledge is non-rival).	China's solar manufacturing scaled globally, reducing costs for everyone. Early RE deployment in Germany/Spain paid the learning cost for the world.	Global learning curve: $c_t = c_0(\sum_{\text{all countries}} Q)^{-\beta}$. Country i 's deployment benefits all countries through lower future costs.	International technology subsidy / coordination. Early deployers should be compensated for the global knowledge spillover (or deployment should be internationally co-financed).	Each country underinvests in deployment because it bears the full cost but captures only a fraction of the learning benefit. Classic public goods underprovision.	Justifies international climate finance for technology deployment. Germany's FiT was a global public good. China's manufacturing scale-up was the single most important climate action of the century. Future equivalent: green H ₂ deployment.
G3	Carbon leakage. Unilateral carbon pricing shifts production to unconstrained jurisdictions, with no net emissions reduction.	Carbon-intensive production relocates from priced to unpriced jurisdictions. Domestic emissions fall; import-embedded emissions rise.	With trade: $E_{\text{global}} = E_{\text{domestic}}(\tau_i) + E_{\text{import}}(\tau_j = 0)$. If import substitution is high, $\partial E_{\text{global}} / \partial \tau_i \approx 0$.	CBAM: equalise carbon cost on imports with domestic price. Eliminates leakage incentive and extends price signal to trading partners.	Without CBAM, unilateral carbon pricing is largely ineffective for traded goods (steel, cement, aluminium, chemicals). With CBAM, it becomes a de facto international carbon price.	EU CBAM is potentially the most impactful international climate instrument. If adopted by EU + UK + US + Japan + Canada (~50% of global imports of CBAM goods), it creates a \$50–100/tCO ₂ floor price on 2–3 GtCO ₂ /yr of industrial emissions globally.

	Assumption / stylised fact	Model in words	Mathematical structure	Optimal international instrument	Deviation from national-only optimum	Practical implication
G4	Fossil fuel subsidy persistence in developing countries. Subsidies are politically entrenched; removal triggers social unrest.	Subsidies keep fossil fuels artificially cheap, creating a large negative carbon price. Removal is politically explosive (Nigeria, Egypt, Ecuador, Indonesia).	Effective carbon price = $\tau_{\text{carbon}} - s_{\text{subsidy}}/e_f$. If $s > \tau \cdot e_f$, the net carbon price is <i>negative</i> even with a carbon tax.	MDB policy-based lending that conditions budget support on gradual subsidy reform with compensation for affected households.	Subsidy reform is the highest-impact domestic policy in many developing countries — larger than any feasible carbon tax. But politically impossible without international budget support and technical assistance.	WB/IMF DPOs with energy reform conditions have driven subsidy reform in Egypt, India, Indonesia. The implied TECP from a single subsidy reform can exceed \$50–100/tCO ₂ — far more than any carbon tax these countries could implement.
G5	Additionality problem in carbon markets. Hard to verify whether credited reductions are “additional” to what would have happened anyway.	Buyer pays for a credit; seller may have reduced emissions anyway. Net effect is buyer relaxes own effort without real global reduction.	If fraction α of credits are additional: net global effect = $(\alpha - 1) \cdot Q_{\text{credits}} \cdot \tau$. For $\alpha < 1$: carbon markets <i>increase</i> global emissions vs no-market counterfactual.	High-integrity standards (ICVCM Core Carbon Principles). Restrict credits to demonstrably additional, permanent reductions. Accept lower volume for higher quality.	Low-integrity carbon markets are worse than no market: they give buyers a license to pollute while delivering fictitious offsets. The VCM’s quality crisis reflects this fundamental problem.	Art 6 design must prioritise additionality over volume. Corresponding adjustments (Art 6.2) are essential. The future of carbon markets depends on whether Art 6.4 can solve what CDM could not.
G6	Climate finance is tiny relative to need. \$100bn/yr committed vs \$1tn/yr needed for EMDE clean energy investment.	International public finance covers ~10% of the investment need. The remaining 90% must come from private capital, which requires derisking.	Total investment needed: $I = I_{\text{public}} + I_{\text{private}}$. Private responds to risk-adjusted returns: $I_{\text{private}}(\text{risk premium, carbon price})$.	Shift climate finance from grants to guarantees. \$1 of guarantee mobilises \$3–10 of private investment; \$1 of grant mobilises \$0.1.	Grant-based climate finance at \$100bn/yr is structurally inadequate. The same money deployed as guarantee capital would mobilise \$300bn–1tn/yr in private investment.	The most important reform in international climate architecture is reorienting MDB balance sheets from lending to guaranteeing. The Bridgetown Agenda, MDB capital adequacy reform, and SDR rechanneling all point here.
G7	Differential climate vulnerability. Countries that contributed least to emissions face the worst impacts and have the least capacity to adapt.	Equity requires transfers from high-emitting rich countries to low-emitting poor countries. But transfers must not undermine mitigation incentives.	Loss and damage $L_i \propto \text{vulnerability}_i \cdot \Delta T$, where $\Delta T \propto \sum_j E_j$. Country i ’s loss caused by country j ’s emissions.	Loss and damage finance + adaptation finance + mitigation finance designed to not create perverse incentives.	Pure compensation (unconditional transfers) doesn’t reduce emissions. Finance must be structured to simultaneously address equity <i>and</i> incentivise mitigation.	CfDs, JETPs, and results-based finance can address both: they compensate developing countries for transition costs while creating mitigation incentives. Pure grants without conditionality address equity but not mitigation.

3. Reading the Tables

Row 0 in each table is the textbook baseline. Each subsequent row adds one realistic complication. Reading down, you accumulate a progressively realistic picture. The key insight is that **every complication adds an instrument to the optimal portfolio** — the Pigouvian carbon tax is never removed, but it's never sufficient alone either.

The national policy stack (Table 1) in order of robustness:

1. Carbon price (corrects the externality) — necessary but not sufficient
2. Technology standards (corrects behavioural myopia + provides political durability)
3. Deployment subsidies (corrects learning spillovers) — can be temporary if tipping point is reached
4. Derisking / concessional finance (corrects capital market failures)
5. Infrastructure investment (corrects coordination failures)
6. Revenue recycling / compensation (manages political economy)

The international policy stack (Table 2) in order of leverage:

1. MDB guarantees and sovereign derisking (highest leverage: \$1 public → \$3–10 private)
2. CBAM (extends carbon pricing internationally at zero fiscal cost)
3. Policy-based lending for subsidy reform (unlocks large implicit carbon price increases)
4. Technology co-financing (corrects global learning spillover)
5. High-integrity carbon markets (direct revenue transfer, if additionality is solved)
6. Grant-based climate finance (lowest leverage but necessary for LDCs and adaptation)

The overarching principle: optimal policy is a portfolio, not a single instrument. Each market failure requires its own instrument (Tinbergen). Each instrument should be chosen for durability given political economy constraints. The TECP measures the combined effect of the portfolio, weighted by actual abatement effectiveness.